Invariant Subspaces and the Jordan Canonical Form

From the class book: Chapter 6, Sections 6.2–6.3

1. Key definitions

Invariant subspace. W is T-invariant if $T(W) \subseteq W$. Kernels and images of polynomials in T, and eigenspaces, are invariant. In a basis extending one of W, $[T] = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$ and $\chi_{T|_W} \mid \chi_T$.

Jordan block. $J_k(\lambda) = \lambda I + N \in M_k(F)$, with 1s just above the diagonal and $N^k = 0 \neq N^{k-1}$. Its characteristic and minimal polynomials are both $(x-\lambda)^k$, and it has one independent eigenvector.

- **Nilpotent:** $N^k = 0$ for some k; the least such k is the index, and $k \leq n$.
- **Triangularizable:** A is similar over F to an upper triangular matrix iff χ_A splits over F; over $\mathbb{C}$ this can be done by a unitary (Schur).

2. Main results

Theorem (Jordan canonical form)

- If χ_A splits, A is similar to $J = \operatorname{diag}(J_{k_1}(\lambda_1), \ldots, J_{k_r}(\lambda_r))$, unique up to the order of blocks. $A \sim B \iff$ same Jordan form.
- For each λ: AM = total size of its blocks; GM = number of blocks; exponent of $(x-\lambda)$ in m_A = largest block.
- With $r_k = \operatorname{rank}(A-\lambda I)^k$, $r_0 = n$: blocks of size $\geq k$ number $r_{k-1} - r_k$; of size exactly k, $r_{k-1} - 2r_k + r_{k+1}$.
- Primary decomposition: if $m_T = \prod p_i^{r_i}$, then $V = \bigoplus \ker p_i(T)^{r_i}$, each summand invariant.

Blocks (one λ, $n=4$)	GM	Exponent in m_A	$\operatorname{rank}(A-\lambda I)$
1,1,1,1	4	1	0
2,1,1	3	2	1
2,2	2	2	2
3,1	2	3	2
4	1	4	3

Counting similarity classes

- Nilpotent $n \times n$ classes $\leftrightarrow$ partitions of n: $p(1), \ldots, p(7) = 1, 2, 3, 5, 7, 11, 15$.
- Classes with $\chi = (x-\lambda)^n$, $m = (x-\lambda)^k$: partitions of n with largest part k. Several eigenvalues: multiply.
- 2×2 and 3×3 matrices are similar iff they have the same χ and m. Every matrix is similar to its transpose.

3. Methods

Method: finding the Jordan form.

- (1) Factor χ_A to get each λ and its AM.
- (2) GM $= n - \operatorname{rank}(A - \lambda I)$ gives the number of blocks; the exponent in m_A gives the largest.
- (3) If that does not fix the sizes (AM ≥ 4), use the ranks r_k of powers.
- (4) Check that the sizes add to n.

Method: powers of a Jordan block.

- $J_k(\lambda)^m = \sum_{j=0}^{k-1} \binom{m}{j} \lambda^{m-j} N^j$; for $k = 2$: $\begin{pmatrix} \lambda^m & m\lambda^{m-1} \\ 0 & \lambda^m \end{pmatrix}$.
- $\operatorname{rank} J_k(0)^j = k - j$ for $j \leq k$.

4. Exam shortcuts and common traps

- If $N^2 = 0$ then $\operatorname{im} N \subseteq \ker N$, so $\operatorname{rank} N \leq n/2$; the Jordan form has $\operatorname{rank} N$ blocks of size 2 and the rest of size 1.
- $J_n(\lambda)$ has exactly $n+1$ invariant subspaces (a chain); $J_n(\lambda)^2$ for $\lambda \neq 0$ is similar to $J_n(\lambda^2)$, but $J_n(0)^2$ splits into blocks of sizes $\lceil n/2 \rceil, \lfloor n/2 \rfloor$.
- A real rotation by $\theta \notin \pi\mathbb{Z}$ has no invariant line in $\mathbb{R}^2$.
- **Traps:** χ and m do not fix the Jordan form for $n \geq 4$ ($(2,2)$ vs $(2,1,1)$); GM is the *number* of blocks, not the largest size; over $\mathbb{R}$ the Jordan form needs χ to split.

5. Solved examples

Example 1. Find the Jordan form of $B = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ -1 & 1 & 2 \end{pmatrix}$.

Solution

- $\chi_B = (x-2)^3$ and $\operatorname{rank}(B - 2I) = 1$, so GM $= 2$: two blocks.
- Sizes $2, 1$: $J = \operatorname{diag}(J_2(2), 2)$ and $m_B = (x-2)^2$.

Example 2. A 7×7 nilpotent N has $\operatorname{rank} N = 4$, $\operatorname{rank} N^2 = 2$, $\operatorname{rank} N^3 = 1$, $N^4 = 0$. Find its block sizes.

Solution

- Blocks of size $\geq 1, \geq 2, \geq 3, \geq 4$: $7-4 = 3$, $4-2 = 2$, $2-1 = 1$, $1-0 = 1$.
- So the sizes are $4, 2, 1$ (sum 7).

Example 3. How many similarity classes have $\chi = (x-2)^4(x-3)^2$ and $m = (x-2)^2(x-3)$?

Solution

- $\lambda = 2$: partitions of 4 with largest part 2: $(2,2), (2,1,1)$. $\lambda = 3$: only $(1,1)$.
- Total $2 \times 1 = 2$.

Example 4. Compute $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}^5$.

Solution

- $J_2(3)^5 = \begin{pmatrix} 3^5 & 5 \cdot 3^4 \\ 0 & 3^5 \end{pmatrix}$.
- $= \begin{pmatrix} 243 & 405 \\ 0 & 243 \end{pmatrix}$.

Example 5. How many similarity classes of 5×5 complex matrices satisfy $N^3 = 0 \neq N^2$?

Solution

- Such N is nilpotent with largest block exactly 3: partitions $(3,2), (3,1,1)$.
- Answer: 2.

Example 6. $\chi_A = (x-1)^2(x-2)^2$ and $m_A = (x-1)(x-2)^2$. Find J, $\dim E_1$ and $\dim E_2$.

Solution

- $\lambda = 1$ has blocks of size 1 only; $\lambda = 2$ has a block of size 2.
- $J = \operatorname{diag}(1, 1, J_2(2))$, $\dim E_1 = 2$, $\dim E_2 = 1$.